

Nicholas Hargreaves, PMP

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Professional Summary

High-impact Sr. Principal Software Engineer and Technical Program Management (TPM) Leader with a distinguished record of executing complex, multi-million dollar cloud and embedded software integrations across global enterprise environments. Expert in architecting scalable connected-vehicle platforms, telemetry services, and high-throughput data pipelines. Proven success navigating major corporate transitions, managing matrixed international teams, and anchoring zero-downtime operations for millions of active endpoints.

Core Competencies

- **Technical & Strategic Leadership:** Enterprise Platform Architecture, Global Team Management (50+ Engineers), Cross-Functional Stakeholder Alignment, Vendor & 3rd Party Management, Budgetary Oversight (\$12M+), Strategic Risk Mitigation.
 - **Systems Architecture & Data Engineering:** Cloud Computing (Azure, AWS, GCP), Microservices, Platform-as-a-Service (PaaS), Distributed Systems, High-Throughput Event Streaming (Kafka, MQTTv5), ClickHouse, Containerization (Docker, Kubernetes/K8s).
 - **Software Engineering & Languages:** Node.js, Electron, Python, C#, C++, C, Rust, SQL, JavaScript, React, JSON, REST APIs, OpenSSL.
 - **Methodologies & Tools:** Agile/Scrum (Certified Scrum Product Owner), Digital Six Sigma (Green Belt), Software FMEA, CI/CD DevOps Pipelines, Jira/Azure DevOps, Enterprise Production SLA Compliance.
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Professional Experience

CARIAD, Inc. (Volkswagen Group) | Redmond, WA

Sr. Principal Software Engineer | Manager, Program Management | 2021 – Present

- **Ensured Production SLA Compliance** for live global MEB and PPE vehicle platforms across the EU and North America, closing the year with **zero open P1/P2 critical production tickets** and preventing platform-induced blockers.
 - **Architected Big Data Health Monitoring Infrastructure:** Migrated a legacy Python script into a production-grade, highly scalable Node.js service backed by ClickHouse and deployed on a Kubernetes (K8s) cluster; enabled scheduled cron-based scanning of **>1M active vehicles per run** tracking **>80 operational attributes**.
 - **Engineered High-Throughput Messaging Pipelines:** Refined GDC message processing workflows to handle production adoption loads exceeding **1,000 messages per second with zero dropped messages**, supporting features like Factory Reset and Digital Twin synchronization.
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- **Applied Statistical Rigor to Incident Resolution:** Led a data-driven investigation into EV Services command acknowledgment failures using production observability tools and MQTT logs; performed statistical analysis ($p < 10^{-2}$ large effect size) comparing EU vs. NA datasets to isolate vehicle session lifecycles as the true root cause, eliminating unnecessary platform changes.
- **Developed Cross-Platform Diagnostics Tooling:** Extended core backend APIs into an accessible, secure, zero-dependency **Electron desktop application** integrated with read-only GITS tenant authentication and compiled into a signed Windows installer for field engineer self-service.
- **Accelerated Global Customer Adoption:** Managed environment readiness and provided critical, on-site/virtual launch engineering support across international manufacturing hubs including Ingolstadt (Europe), Beijing (China), and Chattanooga (North America).
- **Spearheaded Unified Data Environment (UDE) Initiatives:** Designed notification interfaces for VW.AC data integration and engineered a specialized pre-installation utility to preload large-scale vehicle datasets into UDE Kafka pipelines.

Kymeta Corporation, Inc. | Redmond, WA

Director, Program Management | 2016 – 2021

- **Spearheaded the program management office (PMO)** reporting to the C-Suite (COO, CFO, President) to deliver complex land-mobile satellite communications-on-the-move platforms.
- **Controlled an annual program portfolio budget exceeding \$12M**, driving hardware New Product Introduction (NPI), embedded software, and cloud-based OSS/BSS software solutions from quote-to-cash.
- **Established standardized agile tools and delivery processes** that synchronized multiple scrum teams, compressing cycle times and keeping milestones strictly on schedule.
- **Managed high-stakes technical interfaces** between commercial business units, core systems engineering, and strategic partners/customers including SpaceX, OneWeb, and Jaguar Land Rover.

Nuance Communications Inc. | Seattle, WA

Manager, Program Management | Principal Program Manager | 2013 – 2016

- **Built and mentored a senior team** of program managers delivering connected vehicle cloud services to major global automotive OEMs.
- **Owned delivery of head unit and cloud software integrations** for the high-profile Fiat Uconnect In-Vehicle Infotainment (IVI) system, acting as Product Owner within Jira-based Scrum frameworks.
- **Successfully managed aggressive timelines and budgets**, reporting status directly to Senior Accenture management on-site in Turin, Italy.
- **Designed and verified complex backend cloud integrations**, encompassing custom authentication schemes, secure VPN network architectures, and robust end-to-end network security layers.

Tweddle Group Inc. | Bellevue, WA

Senior Program Manager | 2012 – 2013

- **Directed the cross-functional delivery** of head unit applications and connected web services for the Chrysler Uconnect IVI platform, aligning milestones across Sprint and FCA/Chrysler executives.

- **Oversaw a multi-tiered client-server architecture** utilizing Java/Tomcat web servers and Oracle Enterprise databases to manage real-time vehicle provisioning, metrics reporting, and customer billing.
- **Guided UX/UI design, API development, and secure token authorization** for nested head unit software.
- **Negotiated and integrated 3rd-party content feeds** from major external digital vendors including Bing, Yelp, Pandora, and Slacker Radio.

BSQUARE Corp. | Bellevue, WA

Senior Program and Project Manager (Contracted to Ford-Microsoft) | 2006 – 2011

- **Managed a \$13M+ resource and operational budget** for the engineering and quality assurance of middleware applications within the iconic Ford MyFord Touch IVI system.
- **Integrated highly specialized 3rd-party vendor stacks** from Freescale, Audible, TeleNav, Nuance, Broadcom, Gracenote, and Airbiquity under immense delivery pressure.
- **Enforced strict Security Development Lifecycle (SDL)** and systems architecture audits across local and remote development arms.
- **Applied extensive knowledge of vehicle CAN bus networks** to ensure seamless interface compatibility with Bluetooth, USB, iOS, and Android mobile ecosystems.

Senior Program and Project Manager (Professional Services)

- **Owned delivery of Windows CE and Windows Mobile projects** for elite silicon and hardware vendors including Freescale, Dell, Palm, NXP, and Texas Instruments.
- **Drafted and secured formal commercial documents**, including highly technical RFQs, RFPs, and legal Statements of Work (SOWs) for budgets exceeding \$1M.

Motorola Ltd | Swindon, UK

Project Manager / Senior Engineer / Graduate Engineer | 1994 – 2006

- **Program Leadership (2001-2006):** Managed engineering teams (up to 17 heads across the UK and Poland) building UMTS/CDMA 1X Base Station and GSM Base Station Controller firmware. Oversaw staffing budgets up to \$6M and achieved a perfect record with zero non-compliances during formal ISO TL9000 audits.
- **Process Automation:** Authored organizational software policies on automated code generation and utilized Digital Six Sigma methodologies to unlock a 10% reduction in cycle-time via C++ optimizations.
- **Systems & Firmware Engineering (1994-2001):** Designed and implemented POSIX-compliant, IP-based messaging schemes for shared infrastructure code. Led embedded software engineering for GSM BTS Picocellular products from initial equalizing algorithms through to first office application (FOA) field deployments across Sweden, Germany, and the UK.

Professional Certifications & Affiliations

- **Chartered Engineer (CEng)** – Institute of Engineering and Technology (IET), UK

- **Project Management Professional (PMP)** – Project Management Institute (PMI)
 - **Certified Scrum Product Owner (CSPO)** – Scrum Alliance
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Education & Training

- **B.Eng. (Hons) Dip. Eng in Electronic & Communications Engineering** – University of Hull, UK
- **Advanced Technical Certifications:** Software Failure Mode Effect Analysis (FMEA), Digital Six Sigma & Green Belt Training, Design for Six Sigma (Voice of Customer & Critical Parameter Management)
- **Awards:** Kymeta Presidents' Award for Excellence in Program Delivery